

Pokkat

Immersive Technology Development (ITD) Assignment 1 by Mark and Arwen

Project Details

About

Pokkat is an augmented reality (AR) tamagotchi-style cat care game built with Unity and AR Foundation. Players use their device's camera to scan a tracker image, spawning virtual neko (cat) characters into their real-world environment. The game features:

- **AR Image Tracking**
Scan the Pokkat tracker to spawn your main neko into the AR scene
- **AR Plane Detection**
The game detects horizontal surfaces (floors, tables) for realistic grounding
- **Multi-Cat Spawning**
Use multiple physical tracker prints to spawn friend nekos (up to 3 total)
- **Tamagotchi Mechanics**
Hunger and happiness stats that decay over time (0.1 per hour)
- **Interactive Care**
Feed your cat by placing food bowls, pet them, and watch them play with friends
- **Persistent Stats**
Your cat's hunger, happiness, and appearance are saved between sessions. *There exists code that would have been used by Darren to save and load data to a backend like Firebase, but due to how things played out, have been unused.* (`PokkatCore.Statskeeper.LoadFromDict` and `PokkatCore.Statskeeper.SaveToDict`)
- **Time-based Statistic Decay**
While the app is closed, your neko's stats will decay over time. Both hunger and happiness decrease at a rate of **10% per hour** (0.1 per hour). When you reopen the app, the game calculates how many hours have passed since your last session and applies the appropriate decay. If your neko's hunger reaches 0%, the cat dies! Make sure to check in regularly to keep your neko happy and fed.

Instructions

1. **Open the application**, you will be greeted with the main menu:
 - **Start Game:** Brings you to the game screen.
 - **Settings:** Allows you to configure audio volumes.
 - **Exit Game:** Leaves the application.
2. Optionally, tap on the **"Settings"** button to configure audio volumes. Within the Settings page, click on **"Exit"** to leave the menu.

There are three audio sliders you may adjust:
 - **Master Volume:** Controls the overall volume of the game.
 - **Music Volume:** Controls the background music in relation to the master volume.
 - **Sound Effects:** Controls the sound effects volume in relation to the master volume.

▮ Note: Volume uses logarithmic scaling for natural-sounding audio adjustment.
3. Tap on **"Start Game"** to enter the game screen and start the experience.

You will be presented with a view of your device's camera, and a gameplay heads-up display (HUD) that shows two bars:
 - **Hunger Bar:** (Depicted with a food can icon) Indicates the cat's hunger level (0-100%).
 - **Happiness Bar:** (Depicted with a heart icon) Indicates the cat's happiness level (0-100%).

▮ Time Decay: Stats decrease by 10% per hour while the app is closed. If hunger reaches 0%, the cat dies!
4. **Follow the prompt text** displayed on screen:
 - **"Scan the Pokkat tracker to spawn a cat!"** - Point your camera at the Pokkat tracker image to spawn your main neko.
 - **"Move your phone around to detect surfaces!"** - After spawning, slowly pan your phone to detect floor/table surfaces. The neko will fall and land on detected planes.
 - **"Move your phone around to detect more surfaces!"** - The neko needs sufficient plane detection for roaming behaviour.
 - **No prompt text (empty)** - Your neko is happy and the game is in a normal state. You can now interact freely.
5. **Interact with your cat** using the following methods:
 - **Spawning a Food Bowl:** Tap anywhere on a detected surface to place a food bowl. The main neko will walk towards the bowl and eat, restoring **+50% hunger**. Only one bowl can exist at a time; tapping again replaces the bowl.
 - **Petting via Touch:** Tap directly on the neko on your screen. The neko will turn to face you, blink, and bounce happily, restoring **+5% happiness**.
 - **Petting via UI Button:** Tap the "Pet" button on the HUD to pet the main neko (if available).
 - **Spawning Friend Nekos:** Either scan a new tracker at least 25cm away, or remove the existing tracker from view, then move it to a new location (at least 25cm away from any existing nekos) and scan again. A friend neko with a random texture will spawn! You can have up to 3 nekos total.

- **Playing with Friends:** When a friend neko spawns near the main neko, they will turn to face each other and jump together happily, restoring **+50% happiness**.
6. **Satiation State:** Once the cat has been fed (hunger > 50%) and has been petted or played with a friend (happiness > 50%), the game displays: *"The cat is satiated, you can come back again later!"*
 7. At any point, or once the cat is satiated and you need not continue interacting, tap on the **"Exit"** button to leave the game and return to the main menu. Your stats are saved automatically.

Platform and Hardware Requirements

Supported Platforms

Platform	Minimum Version	AR Framework
Android	Android 11.0 (API 30)	ARCore

iOS should technically be supported via ARKit, and I see no reason for my code to not work on iOS, but I did not build for it.

Hardware Requirements

- **Camera:** Device must have a rear-facing camera with AR capabilities
- **Motion Sensors:** Accelerometer and gyroscope required for AR tracking
- **ARCore/ARKit Support:** Device must be certified for ARCore (Android) or support ARKit (iOS)
 - [List of ARCore supported devices](#)
 - ARKit requires iPhone 6s or later, iPad Pro, iPad (5th gen) or later

Tracker Image

The Pokkat tracker image ([Design/TrackingMarker/pokkat-tracking-marker.{jpg,pdf,png}](#)) must be printed or displayed on another screen for scanning. For best results:

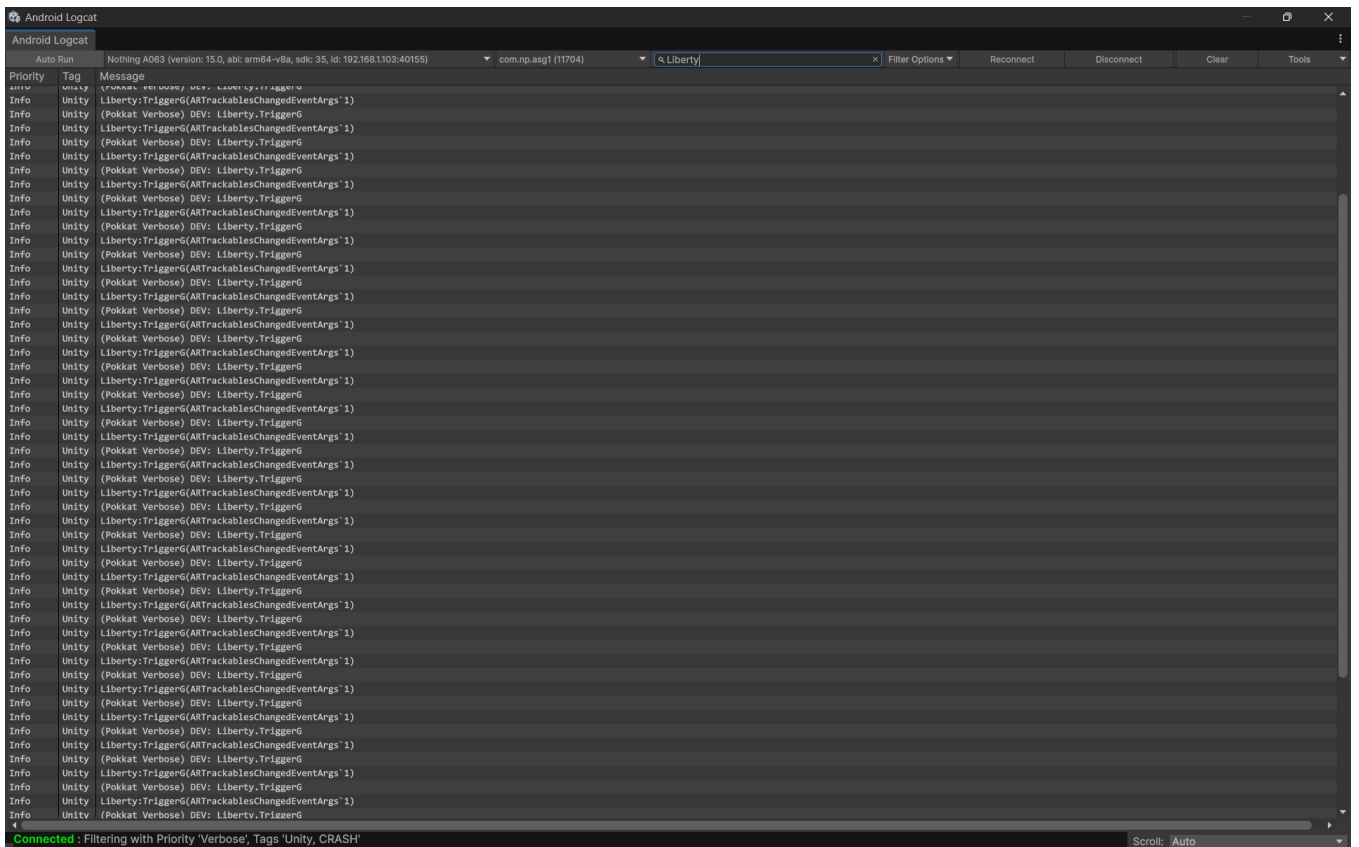
- Print A4-sized
- Use a flat, non-reflective surface
- Ensure good lighting conditions
- Print multiple copies to spawn friend nekos (or bring it in and out of frame)

Limitations and Bugs

Known Limitations

1. **Maximum Nekos:** Only 3 nekos can be spawned simultaneously (1 main + 2 friends). This is deliberate to prevent excessive resource usage and ensure smooth gameplay.
2. **Single Bowl:** Only one food bowl can exist at a time. Placing a new bowl destroys the previous one. This is deliberate.
3. **Idle Roaming Disabled:** The idle roaming behaviour (where nekos walk around randomly) is currently disabled due to edge detection issues with real-world AR plane boundaries. When previously enabled, nekos may walk off detected surfaces.
4. **NavMesh Not Supported:** Unity's NavMesh system does not work reliably with AR Foundation's dynamic plane geometry. The game uses direct AR plane projection instead.
5. **Image Tracking Limitation:** AR systems (ARCore/ARKit) cannot distinguish between multiple physical prints of the same reference image—they share the same trackable ID. The game uses distance-based detection (>25cm apart) to determine if a new neko should spawn.
6. **Plane Drift:** AR planes continuously update their position as tracking refines. Nekos and bowls use XZ-locked anchoring with Y-only stabilisation to prevent horizontal drift.
7. **Simulated Image Tracking May Not Work In Unity Editor/Play Mode:** This is beyond me.

Video: <https://datashower.joshwel.co/public/pokkat/xrsimenv.mp4>
Screenshot of `ARTrackablesEventChangedEventArgs` **being called on an Android build:** <https://datashower.joshwel.co/public/pokkat/Screenshot%202025-12-22%20225926.png>



```
// the code i tried to use to debug this
using System;
using PokkatCore;
using UnityEngine;
using UnityEngine.XR.ARFoundation;

public class Liberty : MonoBehaviour
{
    [SerializeField] private ARTrackedImageManager trackedImageManager;
    private void Awake()
    {
        Logkat.Dev($"Liberty: manager={trackedImageManager}");
    }
    private void OnEnable()
    {
        trackedImageManager.trackablesChanged.AddListener(TriggerG);
    }
    private void OnDisable()
    {
        trackedImageManager.trackablesChanged.RemoveListener(TriggerG);
    }
    private void TriggerG<T>(ARTrackablesChangedEventArgs<T> args) where T : ARTrackable
    {
        Logkat.Dev("Liberty.TriggerG");
    }
    public void Trigger()
    {
        Logkat.Dev("Liberty.Trigger");
    }
}
```

Known Bugs

1. **Spawn Overlap:** Occasionally, AR tracking blips may cause spawn attempts too close to existing entities. A 15cm minimum separation distance is enforced but may not always prevent visual overlap.
2. **Tracking Loss During Fall:** If image tracking is lost while the neko is mid-fall, the neko may land at an unexpected height.

Answer Key / Cheats

- **Quick Friend Spawn:** After the main neko lands, immediately move the tracker to a new spot (>25cm away) and re-scan to spawn friends quickly.
- **Stats Persistence:** Stats are saved in PlayerPrefs. Clearing app data will reset all progress.

- **Respawn Threshold:** Image trackers must be at least 25cm apart to be considered separate neko spawns. Use this to control where friends appear.

References and Credits

Graphical Assets and Models

- **Neko Model and Textures**

Ripped from Neko Atsume Purrfect Kitty Collector, released by Hit-Point Co., Ltd. for the Meta Quest 2+ on 14 December 2023.
<https://www.meta.com/en-gb/experiences/neko-atsume-purrfect-kitty-collector/8401739766534648/>

Ripped by Windows98 on Sketchfab.
<https://sketchfab.com/3d-models/neko-47b59a0fc5084ad88657f43df85a4426>

- **Food Bowl**

Modelled by Arwen!

- **UI Font: One Little Font**

By Konstantina Louka
<https://www.fontspring.com/fonts/konstantina-louka/one-little-font>

Audio and Sounds

- Game/Assets/Pokkat Core Gameplay/Sounds/jump.mp3
"Cartoon Boing Sound Effect 2" by Jessica Hartell on YouTube
<https://www.youtube.com/watch?v=d7vfyFI5kc>
- Game/Assets/Pokkat Core Gameplay/Sounds/eat.mp3
"Nom Nom Nom Sound Effect" by UltraStorm on YouTube
<https://www.youtube.com/watch?v=UaMKUVxidpM>
- Game/Assets/Pokkat Core Gameplay/Sounds/step.mp3
"Wood step Sample 1" by Notarget (Freesound) on Pixabay
<https://pixabay.com/sound-effects/wood-step-sample-1-47664/>
- Game/Assets/Pokkat Core Gameplay/Sounds/bowldown.mp3
"Button push - clicky plastic button" by Gamemaster Audio on Uppbeat
<https://upbeat.io/sfx/button-push-clicky-plastic-button-1/362/4895>
- Game/Assets/Pokkat Core Gameplay/Sounds/pokkat bgm.mp3
"Sweet Cafe | Cute Background Music (Royalty Free)" by Stream Cade on YouTube
<https://www.youtube.com/watch?v=6DhONAQfEVg>
- Game/Assets/Pokkat Core Gameplay/Sounds/meow.mp3
Meow SFX by DRAGON-STUDIO on Pixabay
<https://pixabay.com/sound-effects/meow-sfx-405456/>
- Game/Assets/Pokkat Core Gameplay/Sounds/meow 1 (2).mp3
Cat Meow 1 FX by SOUND_GARAGE on Pixabay
<https://pixabay.com/sound-effects/cat-meow-1-fx-306178/>
- Game/Assets/Pokkat Core Gameplay/Sounds/meow 1 (3).mp3
Cat Meow 7 FX by SOUND_GARAGE on Pixabay
<https://pixabay.com/sound-effects/cat-meow-7-fx-306186/>
- Game/Assets/Pokkat Core Gameplay/Sounds/meow 1 (4).mp3
Cat Meow 8 FX by SOUND_GARAGE on Pixabay
<https://pixabay.com/sound-effects/cat-meow-8-fx-306184/>
- Game/Assets/Pokkat Core Gameplay/Sounds/meow 1 (5).mp3
Cat meow by freesound_community on Pixabay
<https://pixabay.com/sound-effects/cat-meow-85175/>